



PAPER ID-310677

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Subject Code: KCS502

Roll No:



BTECH
(SEM V) THEORY EXAMINATION 2023-24
COMPILER DESIGN

TIME: 3 HRS

M.MARKS: 100

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A**2 x 10 = 20**

Q no.	Question	Marks	CO
1.	Attempt all questions in brief.		
a.	Define bootstrapping in the context of compilers.	2	1
b.	Which phase of compiler is optional and why?	2	1
c.	Explain the concept of shift-reduce parsing.	2	2
d.	Differentiate Parse tree and Syntax tree with an example.	2	2
e.	Define syntax-directed translation schemes.	2	3
f.	What are the two types of attributes that are associated with a grammar symbol?	2	3
g.	Discuss how scope information is represented in a symbol table.	2	4
h.	What is mean by Activation record?	2	4
i.	Discuss two design issues in code generation.	2	5
j.	Explain the concept of global data-flow analysis.	2	5

SECTION B**10x3=30**

2.	Attempt any three of the following:		
a.	Describe the relationship between finite state machines and regular expressions. Discuss how regular expressions are used in lexical analysis and pattern matching.	10	1
b.	For the grammar $S \rightarrow aAd \mid bBd \mid aBe \mid bAe$, $A \rightarrow f$, $B \rightarrow f$ Construct LR(1) Parsing table. Also draw the LALR table.	10	2
c.	Explain the concepts of quadruples and triples in the context of syntax-directed translation. Discuss how they represent intermediate code and support optimization.	10	3
d.	Describe a simple stack allocation scheme for managing memory during program execution. Explain how it is used for storing local variables and managing function calls.	10	4
e.	Explain the role of a code generator in a compiler. Discuss its responsibilities and how it translates intermediate code into the target code.	10	5

SECTION C**10x1=10**

3.	Attempt any one part of the following:		
a.	Given a regular expression $a(b c)^*$, construct the corresponding Nondeterministic Finite Automaton (NFA) that recognizes the language described by the regular expression.	10	1
b.	Check whether given grammar is ambiguous or not. If ambiguous then convert it into unambiguous grammar: $E \rightarrow E+E \mid E^*E \mid id$.	10	1



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10x1=10

4. Attempt any *one* part of the following:

a.	Check whether the given grammar is LR(0) or not: $S \rightarrow PQy$, $P \rightarrow Sy x$, $Q \rightarrow yS$.	10	2
b.	Consider the following grammar for a simple expression language: $E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$ Consider expression $id * (id+id)$. Apply shift reduce parsing to construct parse tree.	10	2

10x1=10

5. Attempt any *one* part of the following:

a.	Write syntax directed definition for a given assignment statement: $S \rightarrow id=E$ $E \rightarrow E+E$ $E \rightarrow E * E$ $E \rightarrow -E$ $E \rightarrow (E)$ $E \rightarrow id$	10	3
b.	Explain how syntax-directed translation handles array references within arithmetic expressions.	10	3

10x1=10

6. Attempt any *one* part of the following:

a.	Define Symbol table? Explain about the data structures used for symbol table.	10	4
b.	Define semantic errors in a compiler. Discuss the challenges associated with detecting and handling semantic errors. Provide examples to illustrate semantic issues.	10	

10x1=10

7. Attempt any *one* part of the following:

a.	Consider a basic block with the following three instructions: 1. $x = a + b$ 2. $y = x * c$ 3. $z = y - d$ Apply common sub expression elimination to optimize the basic block.	10	5
b.	Construct a Directed Acyclic Graph (DAG) to represent the following basic block: 1. $x = a + b$ 2. $y = c - d$ 3. $z = x * y$	10	5